Submission DSA1-MID-SUB06

Question DSA1-MID-Q01
notation changes halfway:
1. 7>4, go right; 7<9, go left; 7>6, go right.
2. Comparison path: 4→9→6; parent of 7 is 6.
marks, subscripts, and units are mixed up
result written as: path 4→9→6; parent=6 (please interpret)

Question DSA1-MID-Q02
notation changes halfway:
1. Maintain a hash set of values seen so far.
2. Before inserting x, if $x \in$ seen return True; otherwise insert x.
3. Expected O(n) time and O(n) auxiliary space; invariant: seen contains exactly the prior prefix.
marks, subscripts, and units are mixed up
result written as: hash set membership scan; expected O(n) time, O(n) space; prefix invariant (please interpret)

Question DSA1-MID-Q03
notation changes halfway:
1. A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.
2. Height is $\Theta(\log n)$, so search is $\Theta(\log n)$.
marks, subscripts, and units are mixed up
result written as: build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree (please interpret)

Question DSA1-MID-Q04
notation changes halfway:
1. Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.
2. Order A,B,C,D,E; layer order equals shortest edge distance.
marks, subscripts, and units are mixed up
result written as: A,B,C,D,E; BFS explores by distance layers and yields fewest edges (please interpret)

Question DSA1-MID-Q05
notation changes halfway:
1. Invariant: the prefix before index i is sorted and contains the original prefix values.
2. Initialization: one-element prefix is sorted; maintenance inserts the next item; termination covers the full array.
marks, subscripts, and units are mixed up
result written as: sorted prefix invariant; initialization, insertion maintenance, termination imply correctness (please interpret)

Question DSA1-MID-Q06
notation changes halfway:
1. 7>4, go right; 7<9, go left; 7>6, go right.
2. Comparison path: 4→9→6; parent of 7 is 6.

marks, subscripts, and units are mixed up
result written as: path 4→9→6; parent=6 (please interpret)

Question DSA1-MID-Q07
notation changes halfway:
1. Maintain a hash set of values seen so far.
2. Before inserting x, if $x \in$ seen return True; otherwise insert x.
3. Expected O(n) time and O(n) auxiliary space; invariant: seen contains exactly the prior prefix.
marks, subscripts, and units are mixed up
result written as: hash set membership scan; expected O(n) time, O(n) space; prefix invariant (please interpret)

Question DSA1-MID-Q08
notation changes halfway:
1. A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.
2. Height is $\Theta(\log n)$, so search is $\Theta(\log n)$.
marks, subscripts, and units are mixed up
result written as: build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree (please interpret)